

NSBM GREEN UNIVERSITY

Faculty of Computing
BSc (Hons) in Software Engineering

Web Application Development – SE102.3 **Model MCQ Examination Paper (Intermediate Level)**

Duration: 60 Minutes

Total Marks: 30

Number of Questions: 30

Instructions to Candidates:

- Answer ALL questions.
- Each question carries ONE (1) mark.
- Select the MOST correct answer for each question.
- Read each code snippet carefully before answering — questions test applied reasoning, not just recall.
- This is a closed-book examination. No electronic devices are permitted.

1. What is the output of the following code?

```
console.log(typeof NaN);
```

- A) number
- B) NaN
- C) undefined
- D) object

2. Given the following CSS rules, what color is applied to an element with class "item" nested inside an element with id="nav"?

```
#nav .item { color: red; }
.item { color: blue; }
```

- A) Red — the ID+class selector has higher specificity
- B) Blue — the later declared rule always wins
- C) Blue — class selectors override ID selectors
- D) It depends on the browser rendering engine

3. Which HTML attribute allows a form field's value to be visible and submitted with the form, but prevents the user from editing it?

- A) disabled
- B) readonly
- C) hidden
- D) inert

4. What is logged to the console?

```
function outer() {
  let count = 0;
  return function () {
    count++;
    return count;
  };
}
const counter = outer();
console.log(counter());
console.log(counter());
```

- A) 1, 1
- B) 1, 2
- C) undefined, undefined
- D) 0, 1

5. What is the output of the following PHP code?

```
$arr = [1, 2, 3, 4];
$result = array_filter($arr, function($n) { return $n % 2 === 0; });
echo implode(",", $result);
```

- A) 1,2,3,4
- B) 2,4
- C) 1,3
- D) Error — array keys must be reset first

6. What is the output of the following code?

```
if ("0") {
```

```

    console.log("A");
  } else {
    console.log("B");
  }

```

- A) A
- B) B
- C) Error
- D) No output

7. A mobile app requires users to complete an 8-field registration form on a small screen. Which approach best reduces cognitive load, according to UX best practice?

- A) Show all 8 fields on one long scrolling screen with clear labels
- B) Break the form into logical multi-step sections with progress indication
- C) Replace labels with placeholder text to save space
- D) Auto-focus every field simultaneously so users can tab freely

8. An element has width: 200px, padding: 20px, border: 5px solid, and box-sizing: content-box. What is its total rendered width?

- A) 200px
- B) 225px
- C) 250px
- D) 245px

9. What is the output of the following PHP code?

```

function calc($a, $b = 10) {
    return $a + $b;
}
echo calc(5) . "-" . calc(5, 2);

```

- A) 15-7
- B) 5-7
- C) 15-2
- D) Error — missing argument

10. What is the output of the following code?

```

console.log([1, 2, 3] + [4, 5, 6]);

```

- A) [1,2,3,4,5,6]
- B) "1,2,34,5,6"
- C) NaN
- D) Error

11. Which best describes a genuine "mobile-first" responsive design approach?

- A) Design for desktop first, then hide elements on mobile with display:none
- B) Write base CSS for small screens, then use min-width media queries to enhance larger screens
- C) Use only max-width media queries throughout the stylesheet
- D) Detect device type with JavaScript and serve different HTML files

12. Which markup correctly associates a label with its input for accessibility, without nesting the input inside the label?

- A) <label>Name</label><input type="text">

- B) <label for="uname">Name</label><input id="uname" type="text">
- C) <input type="text" label="Name">
- D) <label input="uname">Name</label><input id="uname">

13. What is the output of the following PHP code?

```
$x = "10";
$y = 10;
if ($x === $y) {
    echo "Same";
} else {
    echo "Different";
}
```

- A) Same
- B) Different
- C) Error
- D) 10

14. What is logged to the console?

```
for (var i = 0; i < 3; i++) {
    setTimeout(() => console.log(i), 0);
}
```

- A) 0 1 2
- B) 3 3 3
- C) 0 0 0
- D) undefined undefined undefined

15. Which CSS property, when set on a parent element, causes a child with position: absolute to be positioned relative to that parent rather than the viewport?

- A) display: block
- B) position: relative
- C) overflow: hidden
- D) z-index: 1

16. What is the output of the following PHP code?

```
$str = "Hello World";
echo substr($str, -5);
```

- A) Hello
- B) World
- C) Hello
- D) orld

17. What is the output of the following code?

```
let score = 75;
let grade = score >= 90 ? "A" : score >= 75 ? "B" : score >= 60 ? "C" : "F";
console.log(grade);
```

- A) A
- B) B
- C) C
- D) F

18. A navigation menu has 25 top-level links with no grouping. Applying Hick's Law, what change would most improve usability?

- A) Add more links to give users a fuller picture of the site
- B) Group related links into categorized sub-menus to reduce top-level choices
- C) Remove all text labels and use icons only
- D) Randomize the link order on each visit

19. What is the output of the following PHP code?

```
$data = ["a" => 1, "b" => 2, "c" => 3];  
foreach ($data as $key => $value) {  
    if ($value % 2 == 0) continue;  
    echo $key;  
}
```

- A) abc
- B) ac
- C) b
- D) ab

20. A flex container has three items with flex-grow values of 1, 2, and 1 respectively, and 400px of extra space to distribute. How is that space allocated?

- A) Equally — 133px to each item
- B) 100px, 200px, 100px
- C) The first item receives all 400px
- D) Not distributed — flex-grow requires flex-basis to be set

21. What is the output of the following code?

```
const obj = {  
    name: "Alice",  
    greet: function() {  
        return () => "Hi, " + this.name;  
    }  
};  
console.log(obj.greet());
```

- A) Hi, Alice
- B) Hi, undefined
- C) Error
- D) Hi, this.name

22. Which HTML5 input type validates for a correctly formatted address automatically, without any additional JavaScript?

- A) type="text"
- B) type="email"
- C) type="address"
- D) type="mail"

23. What is the output of the following PHP code?

```
$age = 17;  
if ($age >= 18) {  
    echo "Adult";  
} elseif ($age >= 13) {
```

```
    echo "Teen";
} else {
    echo "Child";
}
```

- A) Adult
- B) Teen
- C) Child
- D) Error

24. What is the output of the following code?

```
console.log(0.1 + 0.2 === 0.3);
```

- A) true
- B) false
- C) Error
- D) undefined

25. Which CSS unit is most appropriate for font-size to ensure text scales correctly with a user's browser zoom or accessibility settings?

- A) px
- B) rem
- C) vh
- D) pt

26. What is the output of the following PHP code?

```
echo array_sum(array_map(function($n) { return $n * $n; }, [1, 2, 3]));
```

- A) 6
- B) 14
- C) 36
- D) 9

27. What screen widths does the following media query target?

```
@media (min-width: 768px) and (max-width: 1024px) { }
```

- A) Only screens exactly 768px wide
- B) Screens between 768px and 1024px, inclusive
- C) Screens smaller than 768px
- D) All screens except tablets

28. What is the output of the following code?

```
let x = null;
console.log(x ?? "default");
```

- A) null
- B) default
- C) undefined
- D) Error

29. What is the output of the following PHP code?

```
function isValidUser($role) {
    if ($role == "admin" || $role == "editor") {
        return true;
    }
}
```

```
    return false;  
}  
var_dump(isValidUser("Admin"));
```

- A)** bool(true)
- B)** bool(false)
- C)** NULL
- D)** Error

30. A dashboard shows 15 differently colored buttons, all styled with identical size and font weight, with no visual grouping. Which UX principle is being violated?

- A)** Consistency
- B)** Visual hierarchy
- C)** Feedback
- D)** Affordance

ANSWER KEY

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Q#	Answer	Q#	Answer	Q#	Answer
1	A	11	B	21	A
2	A	12	B	22	B
3	B	13	B	23	B
4	B	14	B	24	B
5	B	15	B	25	B
6	A	16	B	26	B
7	B	17	B	27	B
8	C	18	B	28	B
9	A	19	B	29	B
10	B	20	B	30	B